

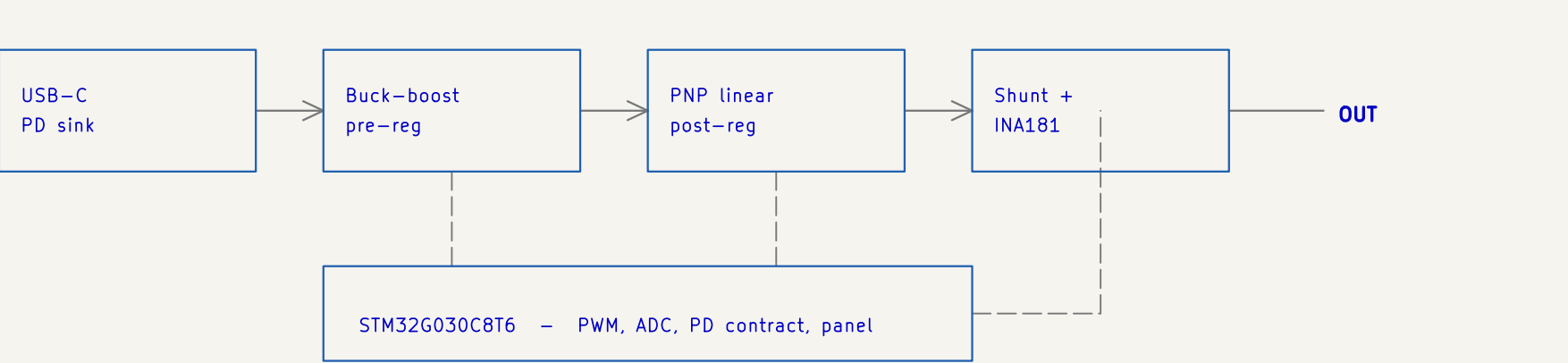
USB-C PD BENCH POWER SUPPLY

0 – 36 V / 0 – 3 A adjustable bench supply from a USB-C PD source

SPECIFICATION

Input	USB-C PD sink, 5 / 9 / 12 / 15 / 20 V contracts
Pre-regulator	4-switch synchronous buck-boost, 250 kHz
Post-regulator	PNP Darlington linear stage, tracking headroom
Output	0 – 36 V, 0 – 3 A
Set resolution	10 mV / 10 mA, PWM reference + 2-pole RC
Readback	voltage and current, 12-bit ADC
Protection	hardware over-current trip, latching
Modes	constant voltage, constant current, battery emulation
Control	STM32G030C8T6, OLED + rotary encoder + 3 keys

BLOCK DIAGRAM



REVISION HISTORY

Rev	MCU	Change
v0.3	CH32V003F4P6	first bench build; NPN Darlington follower
v0.4	CH32V003F4P6	pass device turned round to a PNP
v0.5	CH32V203C8T6	front panel added; the '003 runs out of pins
v0.6	CH32V203C8T6	hardware over-current trip
v1.1	STM32G030C8T6	redrawn as a hierarchy; on-sheet 12 V rail
v1.2	STM32G030C8T6	0–36 V range; analogue nets onto real ADC pins

DESIGNED AND DRAWN BY

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Power Path

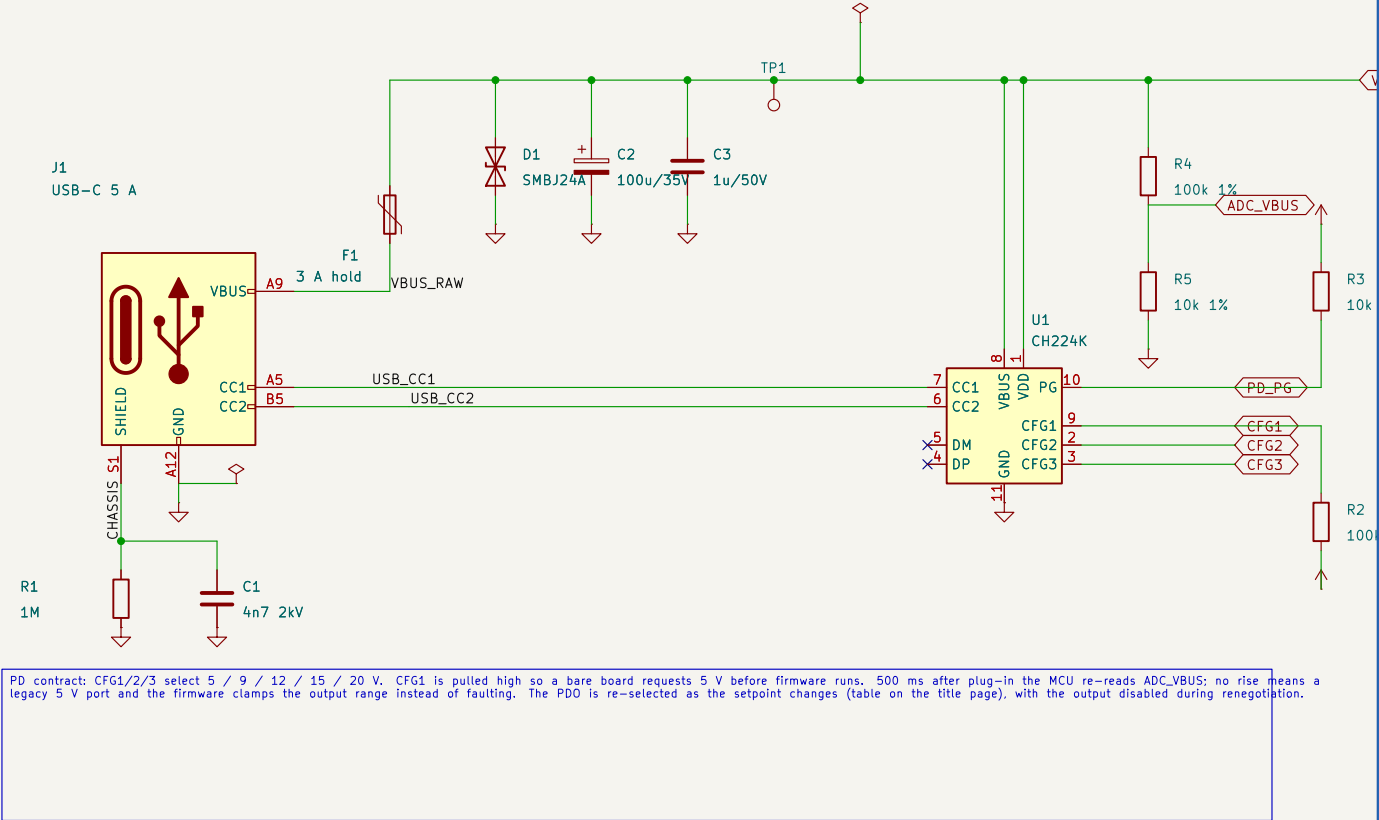
1 input / PD trigger
2 +12 V gate rail
3 buck-boost power stage
4 gate drivers
5 over-current trip

Control and UI

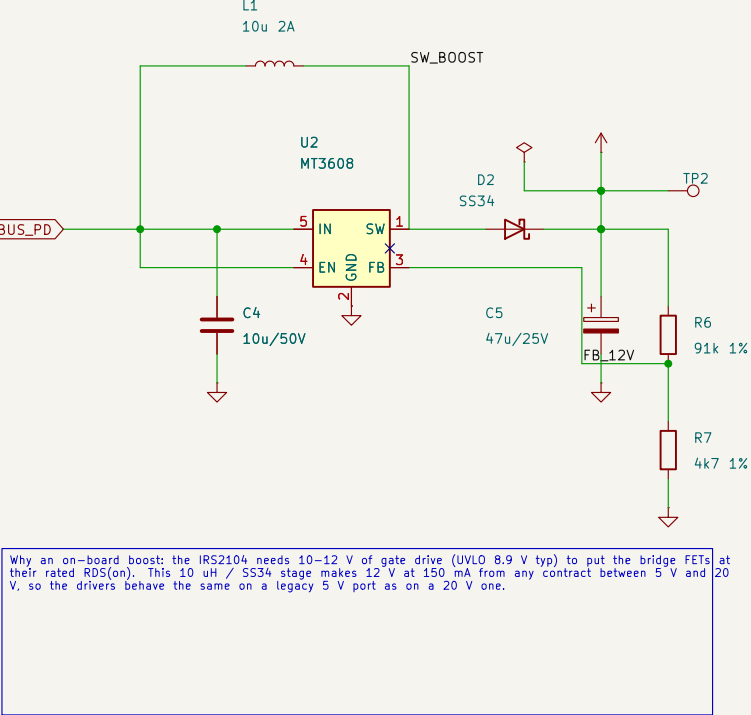
7 linear post-regulator
8 current sense
9 MCU
10 user interface
11 programming header

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Designed and drawn by Mohammadreza Safaeian – m.re.safaeian@gmail.com		
Title page – specification, block diagram, sheet index		
Mohammadreza Safaeian – personal project		
Sheet: /		
File: usbc_pd_psu.kicad_sch		
Title: USB-C PD Bench PSU – 4-Switch Buck-Boost + PNP Linear P		
Size: A3	Date: 2026-04-25	Rev: v1.2
KiCad E.D.A. 10.0.3	Id: 1/3	

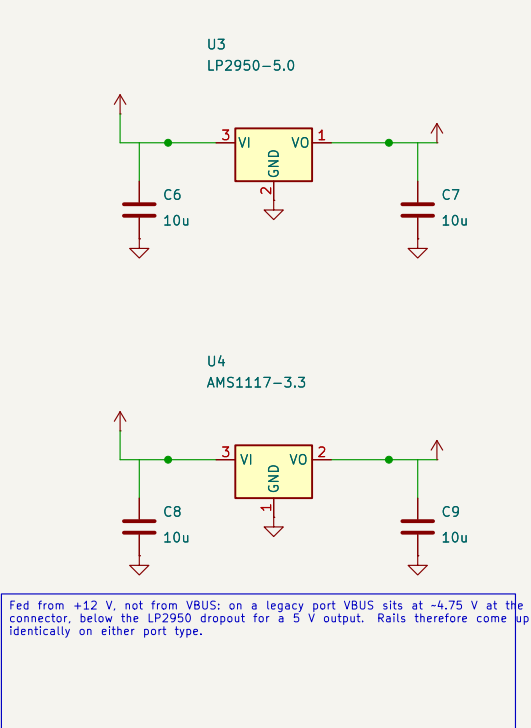
1 - USB-C INPUT, PROTECTION & PD TRIGGER



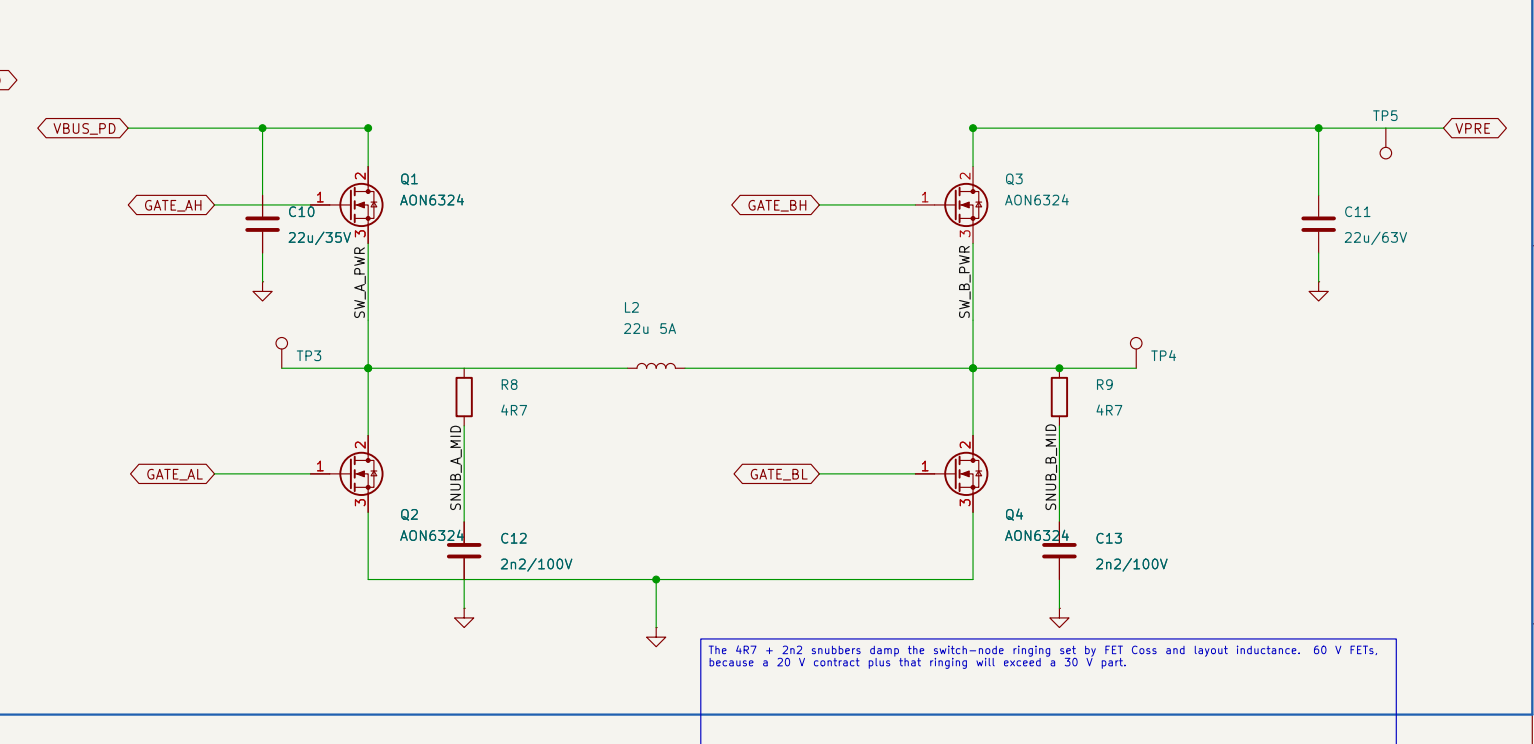
2 - +12 V GATE-DRIVE RAIL (MT3608 boost)



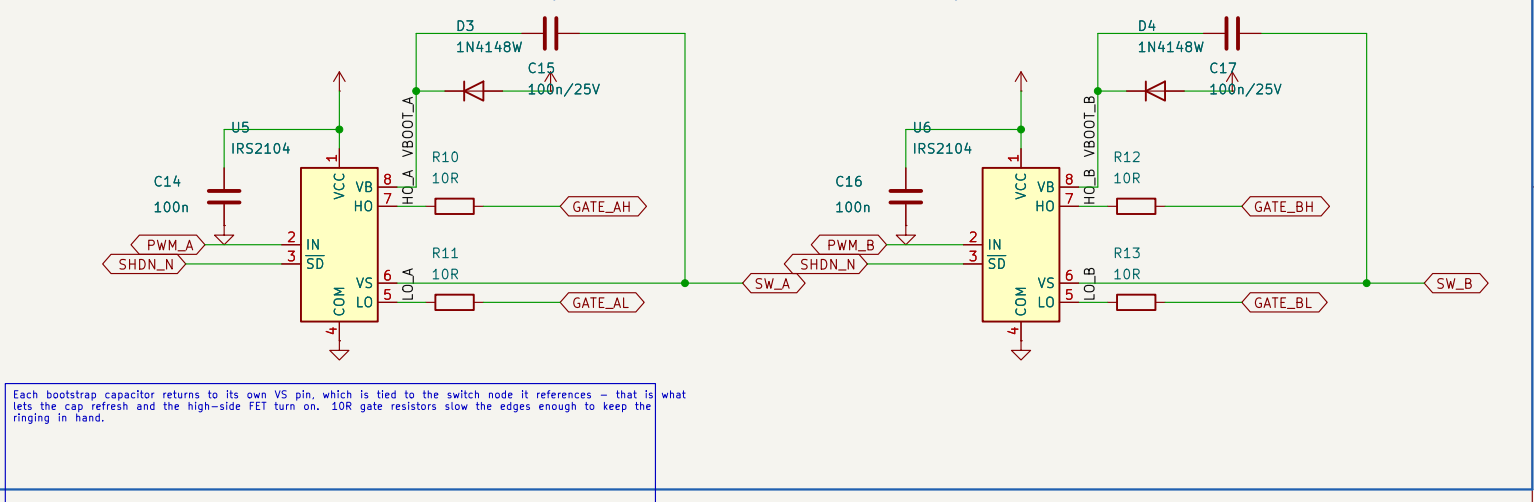
6 - LOGIC SUPPLIES



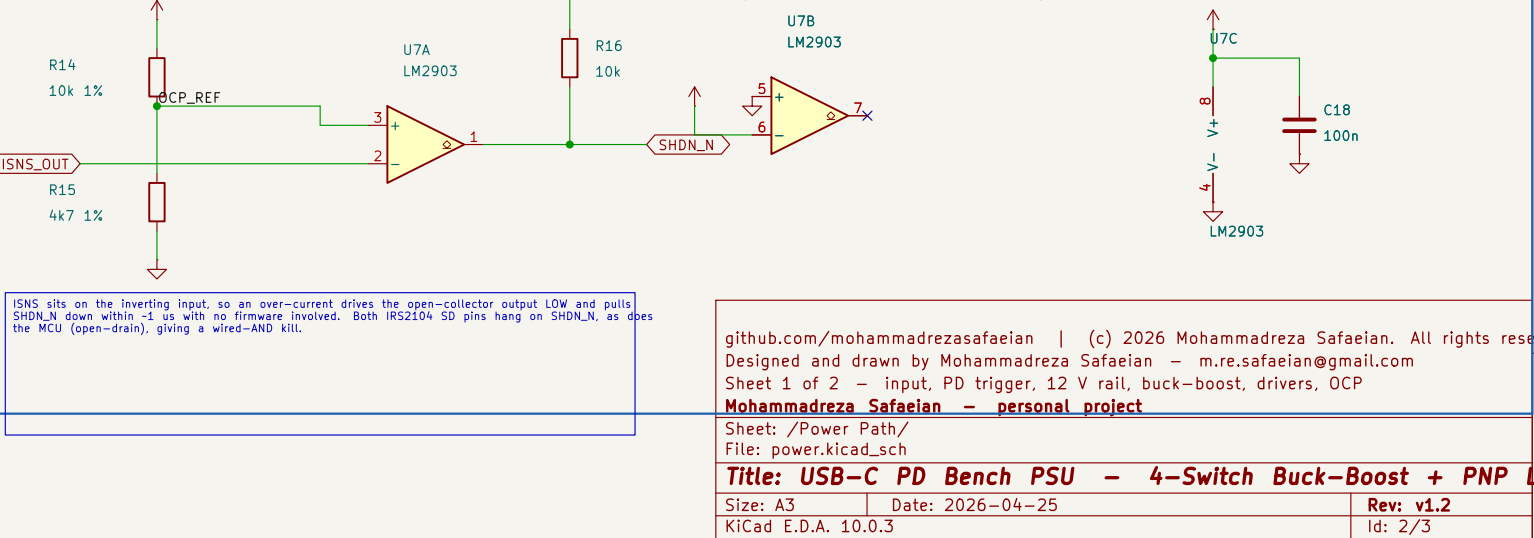
3 - 4-SWITCH SYNCHRONOUS BUCK-BOOST POWER STAGE



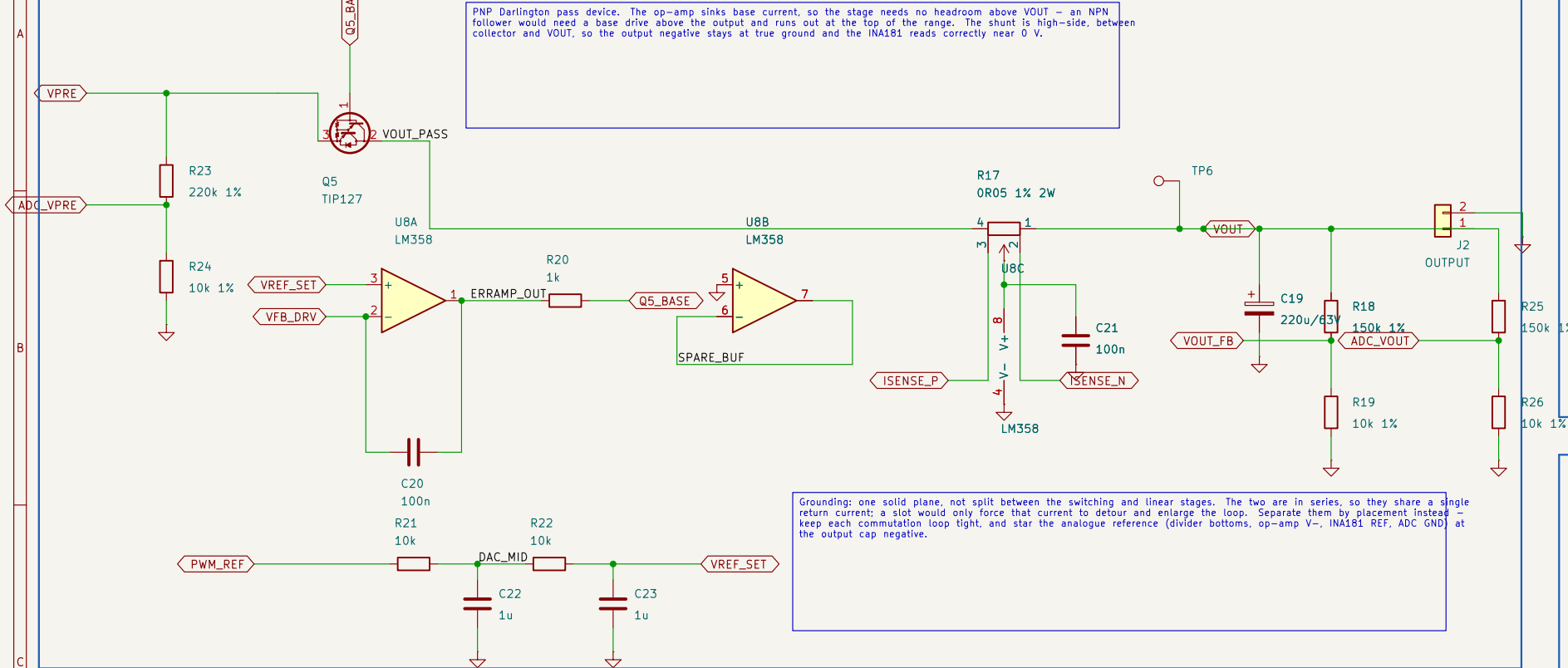
4 - HALF-BRIDGE GATE DRIVERS (IRS2104, 520 ns dead-time)



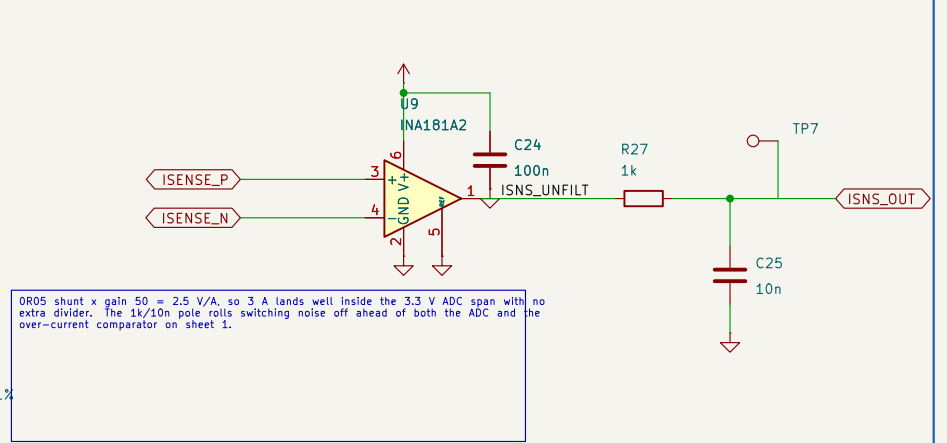
5 - HARDWARE OVER-CURRENT TRIP + LATCH (firmware-independent)



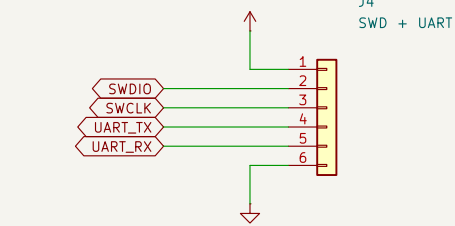
7 - PNP-DARLINGTON LINEAR POST-REGULATOR + OUTPUT



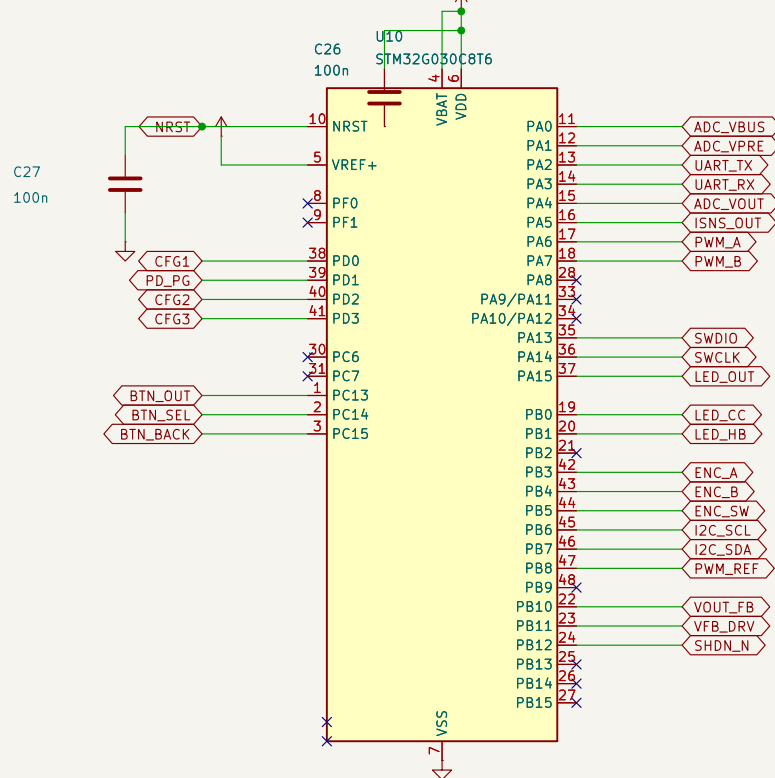
8 - CURRENT-SENSE AMPLIFIER (INA181A2, gain 50)



11 - SWD + UART HEADER



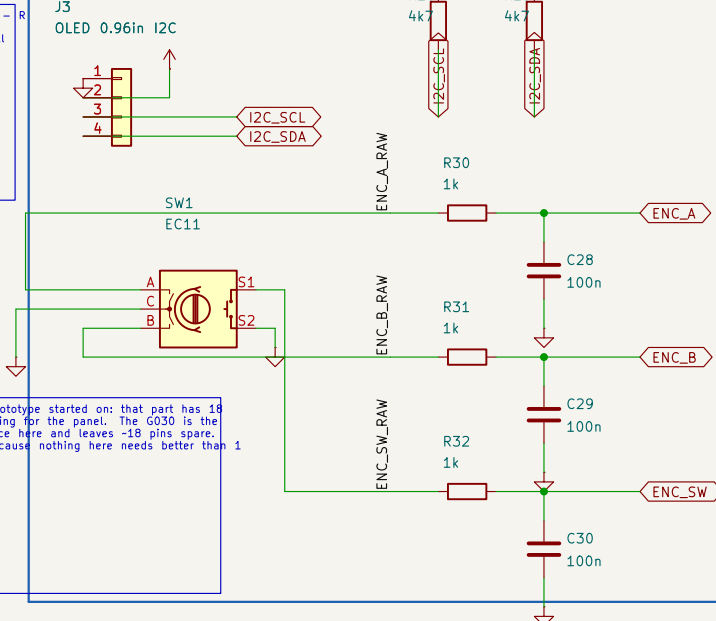
9 - MCU -- STM32G030C8T6 (LQFP-48)



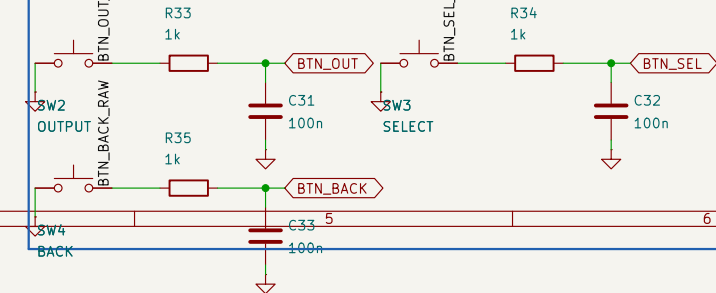
Indicator LEDs sink rather than source: +3V3 - LED -> LED -> GND. PD_PG dark means the board fell back to a legacy 5 V port; SHDN_N lights on an over-current even with the MCU dead.

STM32G030C8T6, chosen over the CH32V003 the prototype started on: that part has 18 GPIO and 17 were already committed, leaving nothing for the panel. The G030 is the cheapest 48-pin Cortex-M0+ that is easy to source here and leaves ~18 pins spare. Runs on the 16 MHz internal RC - no crystal, because nothing here needs better than 1 % timing. SWD on PA13/PA14.

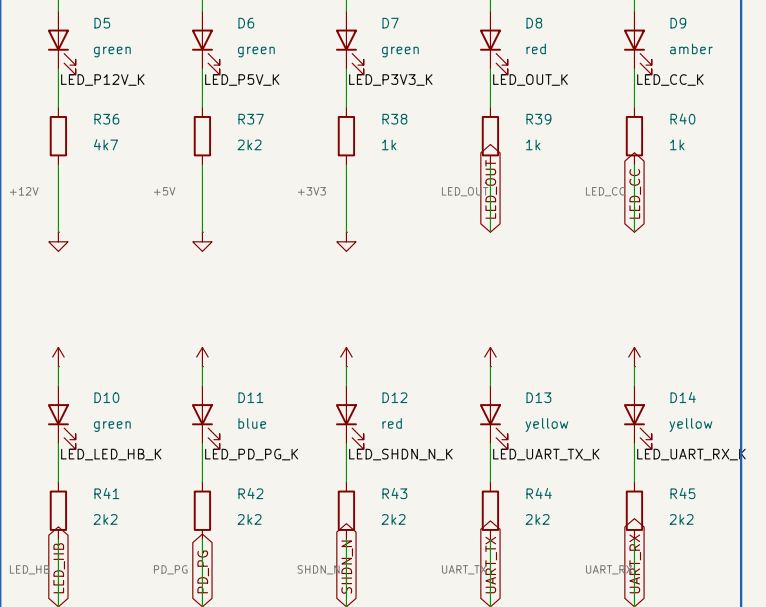
10 - UI: OLED + ROTARY ENCODER



13 - FRONT PANEL



12 - INDICATORS / LED TEST POINTS



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Sheet 2 of 2 - post-regulator, current sense, MCU, UI, programming

Mohammadreza Safaeian - personal project

Sheet: /Control and UI/

File: control.kicad_sch

Title: USB-C PD Bench PSU - 4-Switch Buck-Boost + PNP Linear F

Size: A3

Date: 2026-04-25

Rev: v1.2

KiCad E.D.A. 10.0.3

Id: 3/3